SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security

A1-Threat Intelligence Report

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	5 (all compulsory)	**Marks per Question**	10
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Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

*I **VELUGU KIRAN BOJA RAJU/192312353** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of Student :V.Kiran

Questions :

1. **Threat Hunting Techniques:** An organization suspects that an attacker has gained unauthorized access to its internal network. The security team observes unusual login times, unexpected processes, and connections to unfamiliar external servers. Describe the proactive threat-hunting techniques you would use to investigate the activity and identify potential threats. [10]

2. **SIEM Tools and Security Monitoring :** A company implements a Security Information and Event Management (SIEM) system to monitor its network. The system generates multiple alerts about suspicious login attempts and unusual outbound connections. Explain how the SIEM system can collect and analyze security events, correlate alerts, and help security analysts prioritize potential threats. [10]
3. **Security Monitoring and Anomaly Detection:** A company's security monitoring dashboard identifies a sudden increase in outbound network traffic during non-working hours. The activity originates from a database server that normally transfers only small amounts of data. Explain how you would investigate the anomaly, identify possible data exfiltration, and recommend suitable monitoring and prevention measures. [10]
4. **Tools Used in the IT Industry for Threat Hunting:** An IT organization suspects that malware has infected several employee workstations. The security team has access to SIEM software, endpoint detection tools, network traffic analyzers, and vulnerability scanners. Select suitable tools for investigating the incident and explain how each tool can help identify, analyze, and contain the threat. [10]
5. **Integrated Threat Hunting and Incident Reporting:** A company detects suspicious user logins, connections to a malicious domain, and an unknown process running on an employee workstation. The security team uses Splunk or the ELK Stack to analyze logs and a threat intelligence platform to verify suspicious indicators. Describe the complete threat-hunting process, including data collection, log correlation, threat investigation, malware analysis if required, risk assessment, and preparation of a threat-hunting report with recommended remediation actions. [10]

ANSWERS

1. Threat Hunting Techniques

Threat hunting is a proactive cybersecurity activity in which security analysts search for hidden attackers, malicious behaviour, and indicators of compromise (IoCs) that may have escaped normal security alerts. Unlike purely reactive monitoring, threat hunting begins with a question or hypothesis and uses evidence from endpoints, identity systems, servers, and network devices to test it. In this scenario, unusual login times, unexpected processes, and connections to unfamiliar external servers may be connected signs of unauthorized access.

Establish a Baseline and Define the Hunt

First, establish the normal behaviour of users and systems. Review typical login hours, common source IP addresses, usual processes, expected network destinations, and normal traffic volumes. Compare current events with historical records to identify deviations. Develop a hypothesis such as: “An attacker may be using a compromised account to run an unauthorized process and communicate with an external command-and-control server.” Define the time period, affected systems, and accounts to be investigated.

Investigate Authentication and Identity Logs

Examine Windows Event Logs, Active Directory logs, VPN logs, identity-provider records, and application authentication logs. Look for repeated failed logins, successful logins after many failures, access from unfamiliar locations, logins outside working hours, impossible-travel patterns, new devices, and unusual use of administrator accounts. Check whether multi-factor authentication was used or bypassed. Determine whether the account accessed resources that it does not normally use.

Analyse Endpoint Processes and Persistence

Use an endpoint detection and response (EDR) tool to review running processes, process trees, command-line arguments, loaded modules, file hashes, and recently created files. Pay particular attention to unexpected scripting tools, unsigned executables, suspicious parent-child process relationships, and programs running from temporary or user-writable folders. Check scheduled tasks, startup folders, services, registry run keys, and other persistence locations to determine whether malware can restart after reboot.

Examine Network Traffic and External Connections

Review firewall, proxy, DNS, NetFlow, and packet-capture data. Tools such as Wireshark and Zeek can help identify unusual ports, repeated short connections, suspicious DNS lookups, encrypted sessions to unknown destinations, and large data transfers. Compare destination domains and IP addresses with approved business services and threat-intelligence sources. Repeated connections at regular intervals may indicate command-and-control beaconing, although legitimate software can also produce periodic traffic.

Search for Indicators and Correlate Evidence

Collect indicators such as suspicious IP addresses, domains, URLs, file hashes, filenames, and certificate details. Check them against reliable threat-intelligence feeds and the organization's previous incidents. Search other endpoints and servers for the same indicators to determine whether the incident is isolated or widespread. Correlate events by timestamp, username, hostname, source IP, destination IP, and process ID. A single unusual event may be harmless, but several related events can provide stronger evidence of compromise.

Contain, Document, and Validate

If the evidence indicates an active compromise, follow the organization's incident-response procedure. Isolate affected endpoints where appropriate, disable or reset compromised accounts, block confirmed malicious destinations, and preserve relevant logs and forensic evidence. Document the reason for each action and maintain a timeline. After remediation, continue monitoring for repeated logins, processes, or connections. Avoid deleting files or shutting down systems before evidence is collected unless immediate containment is necessary to protect the organization.

Example: An employee account logs in at 2:00 a.m. from an unfamiliar IP address. Shortly afterwards, a scripting process starts and the workstation connects to a domain flagged by threat intelligence. Correlation of the identity, endpoint, and network records supports a high-priority investigation for possible account compromise and malware activity.

Conclusion: Effective threat hunting combines a clear hypothesis, reliable data collection, baseline comparison, indicator validation, event correlation, and controlled response. The findings should be documented so that detection rules can be improved and similar threats can be identified faster in the future.

2. SIEM Tools and Security Monitoring

A Security Information and Event Management (SIEM) system centrally collects security-related events, makes them searchable, correlates related activities, and generates alerts for investigation. It provides visibility across network devices, endpoints, servers, applications, identity services, and cloud platforms. A SIEM does not automatically prove that an attack has occurred; it helps analysts identify suspicious patterns and decide what needs further investigation.

Data Collection and Normalization

A SIEM collects logs from firewalls, routers, intrusion detection and prevention systems, servers, endpoint protection tools, VPN gateways, databases, DNS servers, authentication systems, and cloud services. Data may be collected through agents, APIs, syslog, or forwarding services. The system parses and normalizes fields such as timestamp, username, hostname, source IP, destination IP, event type, and action. Accurate time synchronization is important because analysts need to reconstruct events in the correct order.

Event Analysis and Detection Rules

The SIEM evaluates events using rules, thresholds, statistical baselines, and sometimes behavioural analytics. Examples include many failed logins from one source, a successful login after repeated failures, access to sensitive files by an unusual account, privilege escalation, malware alerts, or outbound traffic to a suspicious domain. Rules should be tuned to the organization's environment so normal activity does not generate excessive false positives.

Correlation of Multiple Alerts

Correlation connects separate events that share a user, device, IP address, time window, or indicator. For example, a series of failed logins, a successful login from an unfamiliar location, the creation of a new privileged account, and an unusual outbound connection may indicate a single attack sequence. Correlation helps analysts see the larger incident rather than investigating every alert independently. It can also link an endpoint alert with firewall and DNS records.

Alert Prioritization and Risk Assessment

Analysts should prioritize alerts according to severity, confidence, asset criticality, user privileges, threat-intelligence matches, evidence of exploitation, and potential business impact. An alert involving a critical database or administrator account generally deserves more attention than a low-confidence event on a non-critical test system. Analysts should consider whether activity is ongoing, whether sensitive data may be involved, and whether the same indicator appears across multiple systems.

Investigation, Response, and Reporting

When an alert is raised, the analyst reviews the raw events, expands the time window, checks related users and hosts, and builds an event timeline. Threat-intelligence enrichment can help determine whether an IP address or domain is associated with known malicious activity. If the evidence supports an incident, the analyst follows a response playbook, escalates to the incident-response team, and records actions taken. Integrations may support account disabling or IP blocking, but automated actions should be carefully controlled to avoid disrupting legitimate services.

Examples of SIEM Tools

Splunk Enterprise Security provides search, dashboards, correlation searches, and alerting. The ELK Stack—Elasticsearch, Logstash, and Kibana—can ingest, process, search, and visualize logs, with security monitoring capabilities available through Elastic Security. Microsoft Sentinel is a cloud-native SIEM that can collect data from Microsoft and other sources. The best choice depends on scale, budget, data sources, integration needs, and staff expertise.

Example: A SIEM detects repeated VPN login failures followed by a successful login from an unusual location. Five minutes later, the same account accesses a sensitive server and the server initiates an unusual external connection. Correlating these events raises the priority and gives the analyst a timeline for investigation.

Conclusion: A SIEM improves detection by centralizing logs, correlating activity, prioritizing alerts, and supporting incident response. Its effectiveness depends on complete log coverage, well-designed rules, reliable timestamps, protected log storage, and regular tuning.

3. Security Monitoring and Anomaly Detection

Anomaly detection identifies activity that differs from an established baseline. A sudden increase in outbound network traffic from a database server during non-working hours is important because the server normally transfers only small amounts of data. Possible explanations include a legitimate backup, maintenance task, software update, misconfiguration, compromised account, malware activity, or unauthorized data exfiltration. The analyst must collect evidence before deciding which explanation is correct.

Confirm and Characterize the Anomaly

Compare current traffic with historical patterns for the same server, day of the week, and time of day. Record the volume of data sent and received, connection duration, destination, protocol,

port, and frequency. Determine whether the traffic is a single large transfer or many smaller transfers. Check maintenance calendars, backup schedules, replication tasks, and approved data exports to rule out legitimate explanations.

Identify the Destination and Inspect Network Evidence

Use firewall, proxy, DNS, NetFlow, and network detection logs to identify the external IP address or domain and the process or host associated with the connection where available. Check whether the destination is newly observed, has a suspicious reputation, or is unrelated to the organization's business. Wireshark may help inspect packet-level details when a capture is available; Zeek can provide useful connection and DNS metadata. Encrypted traffic may limit payload visibility, so metadata and endpoint logs become especially important.

Review Database and Operating-System Activity

Inspect database audit records to identify queries, exports, table access, unusual account activity, privilege changes, and large result sets. On the server, examine process execution, command lines, newly created archives, compression tools, scheduled tasks, services, and recently modified scripts. Check whether a legitimate backup utility or an unexpected process initiated the transfer. Determine whether the account used had permission to access the data and whether that access was normal.

Determine Whether Data Was Exfiltrated

Data exfiltration is the unauthorized transfer of information from an organization to an external destination. Evidence may include an unusual volume of outbound traffic, sensitive database queries shortly before a transfer, archive creation, connections to suspicious destinations, or use of an account outside its normal pattern. Correlate network events with database audit logs, endpoint telemetry, authentication records, and file access logs. Estimate what data may have been accessed or transferred, but clearly distinguish confirmed evidence from assumptions.

Containment and Evidence Preservation

If exfiltration is strongly suspected, notify the incident-response team and follow the approved response plan. Restrict suspicious outbound connections, block confirmed malicious destinations, or isolate the server if necessary and safe. Consider the operational impact before disrupting a critical database service. Preserve relevant logs, packet captures, database audit records, and forensic artifacts with timestamps and access records. Escalate promptly if regulated or sensitive information may be involved.

Monitoring and Prevention Measures

Implement database activity monitoring and alerts for unusual query volume, bulk exports, off-hours access, and access to sensitive tables. Apply least privilege, separate administrative accounts, enforce multi-factor authentication, patch database software, and regularly review service accounts. Use egress filtering, firewall allowlists where practical, data-loss prevention (DLP), and network segmentation to restrict unnecessary outbound communication. Encrypt sensitive data in transit and at rest, maintain protected audit logs, and test incident-response procedures.

Example: At 1:30 a.m., a database server transfers several gigabytes to a new external host. Audit logs show that a privileged account exported customer records shortly before the transfer. The timing, destination, and database activity together indicate a possible exfiltration incident and justify containment and a detailed investigation.

Conclusion: Anomaly detection is most effective when network monitoring is combined with database auditing, endpoint telemetry, threat intelligence, and clear response procedures. Baselines should be reviewed regularly to account for legitimate changes in business activity.

4. Tools Used in the IT Industry for Threat Hunting

When malware may have infected several employee workstations, analysts should use tools that provide complementary evidence. SIEM systems show events across the organization, EDR tools investigate individual endpoints, network analyzers reveal communication patterns, vulnerability scanners identify weaknesses, and threat-intelligence platforms provide context about suspicious indicators.

SIEM: Splunk or Elastic Security

A SIEM gathers and searches logs from workstations, servers, identity systems, firewalls, and other devices. Analysts can search for malware alerts, repeated authentication failures, unusual process events, suspicious DNS requests, and common indicators across multiple endpoints. Correlation helps establish whether several machines were affected by the same campaign. SIEM evidence is useful for creating a timeline and prioritizing affected assets.

Endpoint Detection and Response (EDR)

Tools such as Microsoft Defender for Endpoint and CrowdStrike Falcon collect endpoint telemetry and detect suspicious process, file, and network behaviour. They can show process ancestry, command lines, file hashes, persistence activity, and connections associated with an

alert. Depending on configuration and permissions, EDR can isolate a device from the network, quarantine a malicious file, or support remote investigation. Analysts should preserve evidence and verify the scope before removing artifacts.

Network Analysis: Wireshark and Zeek

Wireshark is a packet analyzer used to inspect captured network traffic, protocol fields, DNS requests, TCP sessions, and suspicious communication. Zeek monitors network activity and produces structured logs for connections, DNS, HTTP, TLS, and other protocols. These tools can help identify command-and-control patterns, unexpected destinations, repeated beaconing, and possible data exfiltration. They may not reveal the content of properly encrypted traffic, but metadata can still support investigation.

Vulnerability Scanning: Nessus or OpenVAS/Greenbone

Vulnerability scanners identify known weaknesses, missing patches, insecure configurations, and exposed services. Results help determine whether an infected workstation may have been compromised through an unpatched application or vulnerable service. Analysts should validate findings, prioritize critical vulnerabilities, and apply fixes through change management. A vulnerability scan identifies exposure; it does not independently confirm malware infection.

Threat Intelligence and Malware Analysis

A threat-intelligence platform such as MISP or OpenCTI can organize and enrich indicators including IP addresses, domains, URLs, and file hashes. Analysts should review source reliability, confidence, and freshness because indicators can become outdated or shared hosting can cause false positives. For suspicious files, approved malware-analysis tools and isolated virtual machines can help identify behaviour, persistence, and network indicators. Never execute an unknown sample on a production workstation.

Recommended Investigation Workflow

First, use the SIEM to identify affected devices and the earliest suspicious event. Second, investigate each endpoint using EDR and collect process, file, and persistence evidence. Third, review network records with Wireshark or Zeek to identify external communication and possible spread. Fourth, enrich indicators with threat intelligence and use vulnerability scanning to identify weaknesses that need remediation. Fifth, contain confirmed or highly suspected infections, reset affected credentials, remove malware and persistence, patch the entry point, and recover from trusted sources when required.

Example: SIEM alerts show the same suspicious file hash on three workstations. EDR confirms an unfamiliar process and isolates the devices. Network logs show all three contacted the same suspicious domain. Threat intelligence identifies the domain as malicious, while a vulnerability scan finds an outdated application that may have provided an entry point. These combined results guide containment and patching.

Conclusion: No single tool provides the full picture. Combining SIEM, EDR, network analysis, vulnerability scanning, and threat intelligence improves detection, investigation, containment, and prevention.

6. Integrated Threat Hunting and Incident Reporting

Integrated threat hunting combines evidence from identity, endpoint, network, server, and threat-intelligence sources to determine whether suspicious user logins, a malicious-domain connection, and an unknown process are part of the same incident. The investigation should be systematic, evidence-based, repeatable, and documented so that another analyst can understand the reasoning and reproduce the key searches.

Define the Objective and Hypothesis

Start with a clear hypothesis, such as: “An attacker may have obtained a user's credentials, logged in without authorization, executed malware on the workstation, and contacted command-and-control infrastructure.” Define the scope by listing the relevant user accounts, endpoints, time period, critical systems, and business services. Identify what evidence would support or weaken the hypothesis.

Collect and Preserve Relevant Data

Collect authentication and VPN logs, operating-system event logs, EDR telemetry, process and file information, DNS and proxy records, firewall flows, email security alerts, application logs, and relevant cloud audit records. Record where each data source came from and the time range covered. Ensure timestamps are synchronized, preserve original evidence, and restrict access to investigation materials. Missing or incomplete logs should be recorded as a limitation.

Search and Correlate Using Splunk or ELK

Search for unusual logins, failed authentication attempts, new processes, suspicious domains, and outbound connections. Correlate records by username, hostname, source and destination IP, timestamp, process ID, and file hash. Create a timeline showing the initial login, process

execution, domain resolution, outbound connection, and any subsequent activity. Expand the search to other endpoints to see whether the same account or indicators appear elsewhere.

Verify Indicators with Threat Intelligence

Submit suspicious IP addresses, domains, URLs, and hashes to the organization's threat-intelligence platform. Record the source, confidence, context, and date of the intelligence. A known malicious match increases concern, but absence from a feed does not prove that an indicator is safe. Analysts should also check internal allowlists and business context to reduce false positives.

Investigate the Unknown Process and Possible Malware

Inspect the process name, executable path, parent process, command-line arguments, file signature, hash, network connections, and persistence mechanisms. Check whether the process is associated with legitimate software or has suspicious behaviour. If further analysis is required, use an isolated and approved malware-analysis environment. Extract additional indicators where possible and search for them across the environment. Do not execute suspicious files on live systems.

Assess Risk, Impact, and Scope

Determine whether the login was unauthorized, whether the workstation is compromised, and whether other systems or accounts were accessed. Assess the privileges of the affected user, the sensitivity of accessible data, evidence of lateral movement, and the likelihood of continued attacker access. Assign a risk rating based on likelihood, impact, asset criticality, and confidence in the findings. State clearly which conclusions are confirmed, probable, or still uncertain.

Contain, Eradicate, and Recover

Containment may include isolating affected endpoints, disabling or resetting compromised accounts, revoking active sessions or tokens, and blocking confirmed malicious destinations. Preserve evidence before eradication when possible. Remove malware and persistence, patch exploited weaknesses, rotate exposed credentials, and restore systems from trusted backups if necessary. Monitor for repeated indicators and validate that the systems remain clean before closing the incident.

Prepare a Complete Threat-Hunting Report

The report should include: (1) title, date, analyst, and incident identifier; (2) executive summary; (3) objective and scope; (4) data sources and tools used; (5) hunt hypothesis and methodology; (6) timeline of significant events; (7) indicators of compromise; (8) affected

users, devices, and systems; (9) findings and supporting evidence; (10) risk rating and potential business impact; (11) containment and remediation actions; (12) limitations and unresolved questions; and (13) recommendations and follow-up tasks. Include relevant log excerpts or screenshots where permitted, but protect confidential information.

Recommended Remediation and Lessons Learned

Recommended actions include enforcing multi-factor authentication, applying least privilege, patching vulnerable applications, strengthening endpoint protection, restricting unnecessary outbound connections, improving log coverage, and tuning SIEM detection rules. Add confirmed indicators to monitoring controls where appropriate, review account and access policies, and conduct a follow-up hunt to confirm that the attacker has not returned. Lessons learned should be converted into measurable improvements, assigned owners, and target dates.

Example: Logs show an unusual login at 9:12 p.m., an unknown process starting at 9:15 p.m., and a connection to a threat-intelligence-flagged domain at 9:17 p.m. EDR shows the executable was launched from a temporary directory. Correlation across the logs and endpoint evidence supports treating the workstation as potentially compromised. The team preserves evidence, isolates the endpoint, resets the user's credentials, removes the malware, checks other devices for the same indicators, and documents the timeline and remediation in the final report.

Conclusion: An integrated hunt turns separate alerts into a coherent incident narrative. Reliable evidence collection, careful correlation, indicator validation, risk assessment, controlled containment, and a structured report help the organization respond effectively and strengthen future detection.